MILITARY SPECIFICATION SHEET

INTERCOMMUNICATION SETS AN/AIC-14 and AN/AIC-14A

PASSIVE INTERPHONE CONTROL PANEL

C-2642A/AIC-14

Inactive for new design after 10 August 1984
For new design use MIL-I-22352/17(AS) or above.

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the unit described herein shall consist of this document and the latest issue of specification MIL-I-22352(AS).

1. SCOPE

1.1 Scope. This specification sheet provides performance and physical characteristics for the passive Interphone Control panel C-2642A/AIC-14, hereinafter referred to as C-2642A.

1.2 Military Part Number. The military part number shall be M22352/11-1

2. APPLICABLE DOCUMENTS

Refer to 20 of MIL-I-22352(AS) Appendix A for a listing of applicable documents. The requirements specified herein shall have precedence over all referenced documents. The contract and its attachments shall have precedence over any specification or reference document.

3. REQUIREMENTS

3.1 Item Definition. The passive Interphone Control panel shall provide the operator with the capability to activate interconnections with on-board transmitters. The operator using the C-2642A shall have the capability to turn the Interphone Control panel OFF, interact with the associated radio transmitters and have access to interphone signals. The operator shall have the capability of selecting one of eleven transmitters. Audio volume control shall also be provided for the selected receiver input.

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3.1.1 Item Diagram. Passive Interphone Control panel C-2642A shall provide the functions represented schematically in accordance with the diagram of Figure 1.

3.1.2 Interface definition. The passive Interphone Control panel input and output signals will normally display the characteristics presented in Table I.

3.1.3 Controls. The Passive Interphone Control panel shall provide the controls listed in Table II.

TABLE I. Interface signal list.

Signal Name	Characteristics
Inputs:	
Selected Receiver (SEL RCVR)	Voice range audio (300 to 6,000 Hz) signal ranging in amplitude from 0 to 15 Vac. No dc component normally present.
Interphone (INTPH)	Voice range audio (300 to 6,000 Hz) signal ranging in amplitude from 0 to 1 Vac.
Transmit Audio (XMIT AUDIO)	Voice range audio (300 to 6,000 Hz) signal ranging in amplitude from 0 to 1 Vac.
Transmit Key (XMIT KEY)	Relay logic and current sink for lamps having two states: 1) Open and 2) Short to ground
CALL, Emergency Override (CALL)	Relay logic signal: 1) Open 2) Short to ground
External Key (DISC INTPH)	Relay logic signal: 1) Open 2) Short to ground
Disconnect Interphone (DISC INTPH)	Connected to system ground external to the C-2642A
Panel Lights (LIGHTS)	0 to +28Vdc or 0 to 6V, 400 Hz
Power (PWR)	+27.5 ± 0.5 Vdc

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TABLE I. Interface signal list - continued.

Signal Name	Characteristics
Outputs:	
Interphone (INTPH)	Voice range audio (300 to 6,000 Hz) signal ranging in amplitude from 0 to 1 Vac. No dc component normally present.
Selected Receiver (SEL RCVR)	Voice range audio (300 to 6,000 Hz) signal ranging in amplitude from 0 to 15 Vac. No dc component normally present.
Transmit Audio No. 1 (XMIT AUDIO NO. 1)	Voice range audio (300 to 6,000 Hz) signal ranging in amplitude from 0 to 1 Vac. No dc component normally present.
Transmit Audio No. 2 through Transmit Audio No. 11 (XMIT AUDIO NO. X)	Same as (XMIT) Audio No. 1
Transmit Key No. 1 (XMIT KEY NO. 1)	Relay logic: 1) lamp supply voltage 2) short to ground
Transmit Key No. 2 through Transmit Key No. 11 (XMIT KEY NO. X)	Same as Transmit Key No. 1
Interphone Control (INTPH CONTROL)	Relay logic signal: 1) Open 2) Short to ground
Transmit Key (XMIT KEY)	Relay logic signal: 1) Open 2) Short to ground
CALL, Outgo (CALL)	Relay logic signal: 1) greater than 26 Vdc 2) less than 2 Vdc

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TABLE I. Interface signal list - continued.

Signal Name	Characteristics
Grounds:	
Power (PWR GND)	Power return. See 3.4.4.2 of MIL-I-22352(AS).
System (SYS GND)	Signal return. See 3.4.4.3 of MIL-I-22352(AS).

TABLE II. C-2642A/AIC-14 controls list.

Title	Function
TRANS SEL	The C-2642A shall provide a twelve position rotary switch for the selection of one of eleven transmitters for keying and connection of the transmit audio. The first position shall be an ICS position.
ICS ON/OFF	The C-2642A shall provide a two position toggle switch with the following functions: OFF - In the down position the interphone output is disabled. ON - In the up position the system shall select both radio receiver and audio signals from other ICS stations.
RAD VOL	The C-2642A shall provide a control for the continuous adjustment of selected receiver signal output level.

3.1.4 Panel Lamps. The passive Interphone Control panel shall provide two panel lamps to illuminate the front panel.

3.2 Characteristics

3.2.1 Performance Characteristics

3.2.1.1 Input Characteristics

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3.2.1.1.1 Selected Receiver (SEL RCVR). The SEL RCVR signal shall be applied to the RAD VOL control.

3.2.1.1.2 Interphone (INTPH). The INTPH signal shall be applied to the ICS ON/OFF switch.

3.2.1.1.3 Transmit Audio (XMIT AUDIO). The XMIT AUDIO signal shall be applied to the TRANS SEL switches, the output of which will be applied to one of the transmit audio channels No. 1 through No. 11. The XMIT AUDIO input shall display the following characteristics:

3.2.1.1.3.1 Signal Frequency. The XMIT AUDIO input shall accept drive signals which range from 300 to 6,000 Hz.

3.2.1.1.3.2 Signal Amplitude. The XMIT AUDIO input shall accept drive signals which range in amplitude from 0 to 10 Vac.

3.2.1.1.4 Transmit Key (XMIT KEY). The XMIT KEY input shall accept a relay drive logic signal having two states: 1) open and 2) short to ground. The XMIT KEY signal shall be used to produce the XMIT KEY NO. 1 through NO. 11 output on the line associated with the transmitter channel selected on the TRANS SEL control.

3.2.1.1.5 CALL, Emergency Override (CALL). The CALL input shall accept a relay drive logic signal having two states: 1) open and 2) short to ground. The CALL input shall be applied to the Override Relay function which disables the INTPH output of 3.2.1.1.2.

3.2.1.1.6 External Key (EXT KEY). The EXT KEY input shall accept a relay drive logic signal having two states: 1) open and 2) short to ground. The KEY EXT signal shall be applied to the TRANS SEL switch.

3.2.1.1.7 Panel Lights (LIGHTS). The C-2642A shall be capable of accepting power ranging up to +28 Vdc or 6 Vac to illuminate the two front panel lamps. The panel lamps used shall be selected accordingly from TABLE III.

TABLE III. Front panel bulbs.

Nomenclature	Power
MS25237A-327 MS25237A-328	+28 Vdc 6 Vac

3.2.1.1.8 Power (PWR). The C-2642A shall accept and operate within the constraints of this specification when supplied with +27.5 ±0.5 Vdc electrical power as shown in the diagram of Figure 1. The maximum load presented by the passive Interphone Control panel shall not exceed 300 milliamperes.

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3.2.1.2 Output Characteristics. All signals which are passed through this unit from input to output by being routed only through connectors, wire, printed wiring, switches, and relays shall not experience insertion loss in excess of 0.5dB. The C-2642A passive Interphone Control panel shall have the following output characteristics:

3.2.1.2.1 Interphone (INTPH). The INTPH output shall be the same as the INTPH input whenever the ICS ON/OFF switch is ON or when the Override Relay function is activated. When the ICS ON/OFF switch is OFF, the INTPH output shall be an open circuit. The Override Relay shall be energized when the CALL input is a relay logic low (ground).

3.2.1.2.2 Selected Receiver (SEL RCVR). The SEL RCVR output shall be the SEL RCVR input adjusted (reduced) in amplitude by the RAD VOL control. The RAD VOL control shall demonstrate an attenuation range of 30 dB.

3.2.1.2.3 Transmit Audio No. 1 through Transmit Audio No. 11 (XMIT AUDIO NO. X). Only one of the XMIT AUDIO NO. 1 through XMIT AUDIO NO. 11 outputs shall have an active signal at any one time. The specific active line to present the output shall be determined by operator selection of the associated position on the TRANS SEL control.

3.2.1.2.4 Transmit Key No. 1 through Transmit Key No. 11 (XMIT KEY NO. X).
The XMIT KEY output shall be a relay drive logic signal having two states: 1) lamp supply voltage and 2) short to ground. The output shall be a short to ground when the TRANS SEL control is set to the associated channel position and the XMIT KEY input is a ground. Only one channel may be selected for transmitter keying by the TRANS SEL control at any one time.

3.2.1.2.5 Interphone Control (INTPH CONTROL). The INTPH CONTROL shall be taken from the ICS ON/OFF switch. The INTPH CONTROL signal shall be open circuit when the ICS ON/OFF switch is OFF. When ON is selected, the INTPH CONTROL shall be identical to the EXT KEY input. The EXT KEY input shall also be connected to the INTPH CONTROL output through the TRANS SEL switch when ICS is selected.

3.2.1.2.6 Transmit Key (XMIT KEY). The XMIT KEY output shall be taken from the TRANS SEL switch. When the TRANS SEL switch is OFF, the output shall be an open circuit. When the TRANS SEL switch is set to any channel selection, the output shall be identical to the EXT KEY input.

3.2.1.2.7 CALL, Outgo (CALL). The CALL output shall be a relay status sensing line having two states: 1) HI and 2) LO where a high level output exceeds 26 Vdc and a low level output is less than 2 Vdc. The output shall respond to the state of the CALL input in accordance with the following truth table:

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CALL	CALL (OUTGO)
Open	HI
Ground	LO

3.2.2 Physical characteristics.

3.2.2.1 Dimensions. The C-2642A shall consist of a single panel-mounted assembly having the dimensions of Figure 2. All outside dimensions shall be within ±0.03 inches of the nominal value shown.

3.2.2.2 Weight. The C-2642A shall weigh no more than 1.2 pounds.

3.3 DESIGN AND CONSTRUCTION

The design and construction of the passive Interphone Control panel C-2642A shall be in accordance with the requirements of 3.3 of MIL-I-22352(AS).

3.3.1 Front panel lighting. The front panel lamps shall be installed using mounting hardware having red filters in accordance with MIL-L-25467 to enhance night vision. The panel lamps shall be mounted in accordance with the outline drawing Figure 2. The lamp drive circuitry shall be capable of accommodating power ranging up to 28 Vdc or 6 V, 400 Hz ac.

4. QUALITY ASSURANCE PROVISIONS

See MIL-I-22352(AS).

4.1 Responsibility for tests and inspections. See MIL-I-22352(AS).

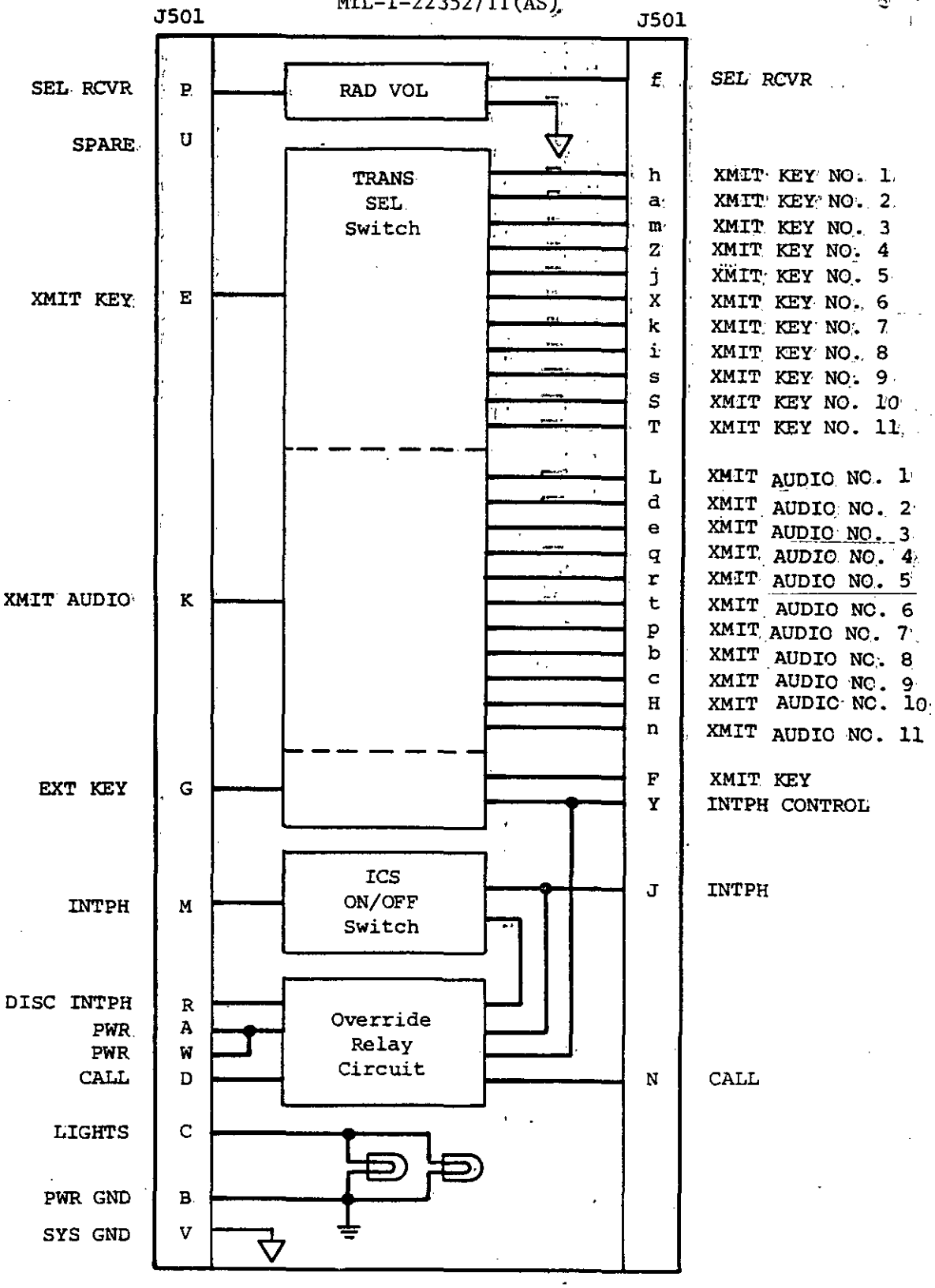
4.2 Test plan. A detailed test plan will be submitted by the contractor for approval by the acquiring activity which details the tests to be performed in order to verify compliance with the performance characteristics specified in 3.2.1. The test plan will consider all of the inspections delineated in 4.2 of MIL-I-22352(AS).

4.3 Test procedures. Detailed test procedures will be submitted by the contractor for approval by the acquiring activity which will satisfy the requirements of the test plan of 4.2.

Preparing Activity:
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Project No. 5831-N042-11

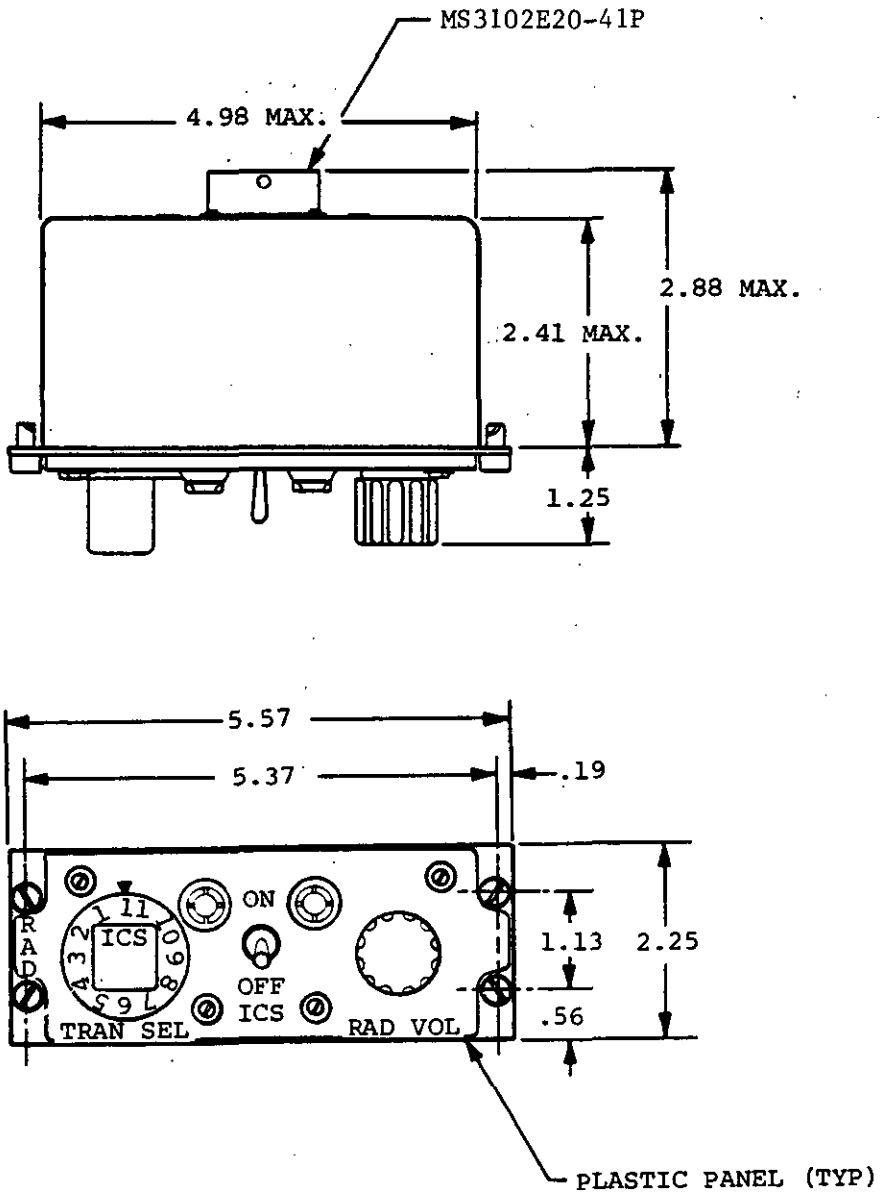
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1/ Disconnect Interphone is connected to ground external to the C-2642A.

FIGURE 1. C-2642A functional diagram.

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1/ All dimensions are in inches. Tolerances on dimensions are ±0.03 inches unless otherwise specified.

FIGURE 2. C-2642A outline drawing.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-22352/11(AS)	INTERCOMMUNICATION SETS, AN/AIC-14 AND AN/AIC-14A PASSIVE INTERPHONE CONTROL PANEL C-2642A/AIC-14

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

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6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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